Introduction

Modern life relies heavily on digital technology, from online banking and smartphones to cloud platforms and connected devices. As organisations grow more dependent on these systems, attackers gain more opportunities to exploit weaknesses in digital infrastructure. Reports show that increasing digitalisation has widened the attack surface available to cybercriminals, allowing them to carry out more frequent and sophisticated operations (ENISA, 2023). Cybersecurity aims to protect confidentiality, integrity and availability, but because technology develops quickly and new attack methods continue to appear, security has to be treated as something ongoing rather than a single solution.

Cultural institutions face particular challenges in this landscape. Their weaknesses often stem from a mixture of outdated systems, inconsistent internal processes and human behaviour. Research indicates that a large number of security incidents are caused by simple mistakes, meaning that everyday actions such as responding to a deceptive email or incorrectly configuring a system can have serious consequences (IBM Security, 2021). Many cultural organisations depend on ageing catalogues, unsupported software and public facing services, all of which expand the attack surface and increase exposure. When these issues are combined with limited security resources and uneven awareness among staff, cultural institutions become attractive targets for attackers.

The British Library ransomware attack in October 2023 illustrates how these risks come together in practice. The Rhysida group carried out a double extortion attack that involved both data theft and system encryption, leading to prolonged disruption across the Library's digital services. This case provides a useful example of how technical weaknesses, human factors and organisational limitations can interact to produce a serious incident. It also reflects wider challenges faced by cultural heritage organisations, where long term underinvestment and legacy technology create systemic vulnerabilities. Analysing the attack therefore offers insight into how adversaries exploit structural weaknesses and how cyber incidents can affect public access to knowledge and cultural infrastructure.

Cyber Incident Scenario Analysis

Motivations of the attack

The motivations behind the British Library attack resemble those seen in many modern ransomware cases. Rhysida is financially motivated and uses a double extortion approach by stealing data before encrypting systems. Connolly and Morgan (2020) note that this tactic has become more common because it increases pressure on victims by combining operational disruption with the threat of leaked information. Cultural institutions often hold valuable personal data but have limited security budgets, making them appealing targets where attackers expect weaker resistance.

The attack also shows elements of opportunism. Many public institutions rely on outdated technology and inconsistent processes, creating weaknesses that criminals can identify through routine scanning. Research by Mersinas, Liu and Panteli (2024) describes ransomware groups as organised operations that choose targets based on visibility, potential disruption and the chance of payment. While Rhysida does not appear to be politically motivated, attacking a national institution raises the group's profile within criminal networks. Their behaviour reflects a professionalised model in which reputation and disruption provide leverage. This differs from politically driven campaigns that focus on ideological or strategic objectives. The choice of a high profile but resource limited institution suggests a calculated decision to maximise impact while facing minimal resistance.

Attack vector, attribution, assets targeted and security objectives violated

Although the British Library did not release detailed technical information about the initial breach, Rhysida's known methods point to several likely entry routes. These include phishing emails, credential harvesting and exploiting remote access services that were misconfigured or not patched. Attackers often use automated tools to identify exposed systems, and phishing continues to be effective because it takes advantage of predictable human behaviour.

Attribution is straightforward because Rhysida publicly claimed responsibility and used techniques similar to those seen in previous attacks. Once inside the network, the attackers moved laterally and escalated their access. They reached key digital assets

such as catalogue servers, HR databases and internal repositories. These systems contain essential organisational information, technological infrastructure and sensitive personal data.

Several security objectives were compromised. Confidentiality was lost when staff data and internal documents were stolen and later published. Integrity was affected through large scale file encryption that made data unusable. Availability suffered because major digital services remained offline for extended periods. Authenticity and accountability were also weakened because damaged or incomplete logs made it difficult to verify events during the early investigation. This shows how one incident can disrupt multiple layers of security at the same time.

The attack can also be understood through the Cyber Kill Chain. Phishing or credential theft fits the reconnaissance and initial access stages. Lateral movement and privilege escalation reflect the expansion of control within the environment. Data exfiltration and encryption represent the final actions on objectives. This sequence shows how the attackers progressed through several stages without detection.

Vulnerabilities exploited

The success of the attack resulted from a combination of technical, human and organisational vulnerabilities. Many cultural institutions still rely on legacy systems that lack modern security features. Holmström (2025) found that outdated infrastructure increases the likelihood of attack and the severity of the outcome because it is more difficult to secure and maintain. Older catalogue platforms and authentication systems often use configurations that are no longer considered secure, making them easier for attackers to exploit.

Human behaviour also played a part. Kadena (2021) notes that many breaches begin with simple user mistakes such as clicking a malicious link or reusing a weak password. Attackers design their techniques around familiar behaviours including trust in internal communications and rapid responses to urgent messages.

Organisational weaknesses contributed further. Slow patching cycles, fragmented IT governance and limited monitoring abilities allowed attackers to explore the network without being noticed. Poor network segmentation likely made it easier for the ransomware to spread between systems, which aligns with Connolly and Morgan's (2020) findings about how segmentation failures intensify ransomware impact.

These vulnerabilities did not develop independently. They reflect long standing issues across the public sector, where restricted funding, slow procurement processes and reliance on bespoke systems make it difficult to introduce modern security measures. This creates environments in which outdated configurations persist for long periods, increasing the risk of a successful intrusion.

Consequences and extended impacts

The consequences of the attack were severe. The British Library lost access to core digital services including its online catalogue, website functions and internal systems. Recovery required extensive rebuilding work that continued for several months. The theft of staff information also created lasting risks such as identity fraud and personal distress.

The impact was felt far beyond the Library. Students, researchers and partner organisations experienced interruptions to their work, highlighting how dependent the wider community has become on digital cultural infrastructure. Holmström's (2025) study of similar public sector incidents shows that the effects of ransomware often reach entire communities rather than remaining confined to the affected organisation. The attack also raised wider concerns about the resilience of cultural institutions with ageing systems and limited budgets. It exposed structural weaknesses and contributed to reduced trust in public digital services, underlining the fragility of the cultural sector when facing modern cyber threats.

Lessons learned

There are several lessons that cultural and public institutions can draw from this incident. The first concerns legacy technology. Many organisations depend on outdated authentication systems, catalogue software and infrastructure. Holmström (2025) argues that such environments make both compromise and recovery more difficult. Modernising these systems or applying strong compensating controls should therefore be a priority.

A second lesson involves human behaviour. As Kadena (2021) explains, attackers exploit predictable user habits. Organisations with varied workforces, including volunteers and temporary staff, face uneven security practices. Improving awareness, strengthening authentication and creating clear reporting routes can help reduce risk.

A third lesson relates to internal governance. Fragmented structures, irregular patching routines and limited monitoring capacity allowed attackers to move through the Library's systems without detection. Connolly and Morgan (2020) note that weak oversight and poor segmentation increase exposure to ransomware. More systematic risk assessments, clearer responsibilities and stronger access controls would help address these issues.

Transparency and accountability also matter. The publication of staff data caused distress and highlighted the need for clear communication and support following a breach.

Finally, the incident showed the importance of sector wide resilience. Similar institutions often face the same financial pressures and technological constraints, meaning that disruptions can have national or international consequences. Cybersecurity depends on people, processes and technology all working effectively together, and weaknesses in any of these areas can lead to significant harm.

Although each lesson is important, the most urgent priorities are improving internal governance and modernising legacy systems. Legacy environments amplify other weaknesses, while poor governance prevents organisations from applying consistent protective measures. Addressing these challenges requires coordinated investment across the sector.

Application of cyber security tools and techniques

A range of cyber security tools and techniques could have reduced the likelihood or impact of the British Library attack. Preventative measures form the first line of defence. Multi factor authentication reduces the usefulness of stolen credentials by requiring a second verification step. Regular vulnerability scanning, configuration checks and timely patching help identify weaknesses before attackers discover them.

Network segmentation is also essential. Tools such as firewalls, virtual LANs and access control lists divide systems into controlled zones. This limits the spread of malware by preventing attackers from moving easily between unrelated parts of the network.

Detection tools add another vital layer. Intrusion detection systems, behavioural monitoring and centralised logging can highlight unusual activity that suggests a

breach. Tools that analyse traffic patterns can detect suspicious connections, while file integrity monitoring can reveal early signs of ransomware. Endpoint detection and response tools can isolate infected devices, preventing wider damage.

Recovery tools are crucial for resilience. Automated, secure backups allow organisations to restore data even if large portions of the network are encrypted. Effective backups should include multiple versions and offsite storage, along with regular testing to ensure they can be used during an emergency.

Risk assessment and threat modelling tools help organisations understand likely attack paths and prioritise their resources. Mapping assets and possible vulnerabilities enables more informed decisions about which protective measures to implement.

These tools are most effective when used together. Multi factor authentication reduces the chance of initial compromise. Segmentation restricts movement within the network. Monitoring tools highlight suspicious behaviour, and backups support recovery. When combined, these measures create a layered defence that addresses the weaknesses observed in the British Library incident.

Conclusion

The British Library ransomware attack shows how vulnerabilities across technology, behaviour and organisational processes can combine to create widespread disruption. Financially motivated groups such as Rhysida exploit outdated systems, inconsistent practices and predictable human behaviour to gain access and cause serious damage. The effects continued well beyond the Library itself, interrupting research activity and limiting access to cultural resources.

To strengthen resilience, organisations should update legacy systems, improve authentication and review permissions regularly. Stronger monitoring and early detection are essential for reducing the spread of future attacks. Recovery planning, including secure and tested backups, is equally important.

Broader improvements are also needed. Organisations should establish clearer governance structures, strengthen accountability and continue to develop staff awareness. Regular risk assessments and appropriate training can reduce human error and improve incident response.

Overall, the attack highlights the need for ongoing improvement and sustained investment in cybersecurity. Institutions facing similar constraints can reduce their

exposure by addressing weaknesses across people, processes and technology. Looking ahead, cultural organisations will require adaptive security strategies that evolve with the threat landscape. Long term resilience depends on coordinated investment and recognition that cybersecurity is now fundamental to providing reliable public access to cultural services.

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